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Divergent climate impacts of alternative feed ingredients in Chinese pig production

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Food systems account for roughly one-third of global greenhouse gas emissions, with China's pig sector alone contributing an estimated 120–130 Mt. CO₂-eq annually. Feed production dominates this footprint, yet evidence comparing alternative ingredients under Chinese conditions is limited by inconsistent methodology and the frequent omission of waste valorization credits. This study presents a cradle-to-farm gate life cycle assessment of four feeding scenarios for Chinese growing-finishing pigs: conventional corn-soybean meal, black soldier fly larvae (BSFL) meal at 10 and 20% dietary inclusion, and 30% processed food waste. Tier 1 manure emission factors and ISO 14044 system expansion for avoided landfill emissions were applied. The conventional baseline produced 3.9 kg CO₂-eq per kg live weight, consistent with recent Chinese benchmarks. Food waste feeding reduced this by 28% to 2.8 kg CO₂-eq/kg LW, driven by avoided maize production and a landfill methane credit. BSFL meal increased emissions by 31–36% above baseline due to energy-intensive processing at pilot scale, a burden not offset by co-product allocation alone. Monte Carlo uncertainty analysis (1,000 iterations) yielded non-overlapping 95% confidence intervals across all four scenarios, confirming that rank-order conclusions are robust to the full parameter space tested. Food waste feeding is near-term deployable within existing infrastructure, contingent on harmonized biosecurity regulation; BSFL meal requires energy system decarbonization to become climate-competitive, as a 30% reduction in production energy narrows the gap with baseline by only 10–11% under China's current grid.

KEYWORDS

carbon footprint, circular economy, *Hermetia illucens*, life cycle assessment, swine feed substitution

1 Introduction

Food systems are responsible for approximately one-third of global anthropogenic greenhouse gas (GHG) emissions, with livestock production contributing a substantial share (Food and Agriculture Organization of the United Nations, 2024; Crippa et al., 2021). Kim et al. (2024) report that pork accounted for approximately 33–37% of global meat consumption in 2022, confirming its position as a major contributor to the global supply. Thus, as demand for animal-source protein continues to grow, mitigating the environmental footprint of pig production has become central to sustainable food system transformation.

China plays a pivotal role in this transition. As the world's largest pork producer and consumer, China accounts for approximately half of global pig production (Food and Agriculture Organization of the United Nations, 2024; Bai et al., 2018). The sector consumes over 180 million tonnes of feed annually, with corn and soybean meal constituting the predominant

ingredients in commercial pig diets (Steinfeld et al., 2006). However, China imports the majority of its soybean requirements, creating significant trade vulnerabilities and agricultural dependencies; this import reliance, coupled with rising environmental concerns about intensive livestock production, has prompted policy initiatives to diversify protein sources (Bai et al., 2018). Life cycle assessments estimate that China's pork sector emits approximately 120–130 Mt. CO₂-eq annually, with average emission intensities around 3.5–4.0 kg CO₂-eq per kg live weight (Fan et al., 2023; Chen et al., 2021). Provincial-level analyses further highlight significant spatial heterogeneity in production efficiency and carbon intensity (Fan et al., 2023). Across production systems, feed cultivation, processing, and transport consistently emerge as the dominant contributors to life-cycle emissions (Poore and Nemecek, 2018; Yang et al., 2023).

As a result, recognition of feed production's environmental importance has catalyzed exploration of alternative protein sources. For instance, Black soldier fly larvae (BSFL; *Hermetia illucens*) production converts organic waste streams into high-quality protein while generating marketable co-products including insect oil and organic fertilizer frass (van Huis et al., 2013). Similarly, processed food waste feeding practiced extensively in several Asian economies diverts edible residues from landfill disposal while maintaining nutritional adequacy for pig production (Zu Ermgassen et al., 2016). Both approaches align with waste reduction and resource efficiency objectives, yet their comparative environmental performance within Chinese production systems remains incompletely quantified. Generally, three categories have gained attention: insect-based feeds, food waste valorization, and alternative plant proteins. Insect-based feeds, particularly BSFL, have been studied for their potential to upcycle organic waste streams into protein (Smetana et al., 2019). BSFL meal contains 40–60% protein with amino acid profiles suitable for monogastric animals (van Huis et al., 2013). Life cycle assessment studies demonstrate that environmental performance depends critically on production system design, particularly energy requirements for climate control and processing (Smetana et al., 2019; Halloran et al., 2016).

Insect-based feeds have attracted particular LCA scrutiny because of their energy-intensive production requirements. Published life cycle assessments of BSFL meal production report global warming potential values ranging from 1.4 to 5.8 kg CO₂-eq per kg protein, a fivefold spread that reflects differences in energy sources, rearing substrate composition, and system boundary definitions across studies (Smetana et al., 2019; Halloran et al., 2016). Energy inputs for heating, ventilation, and mechanical processing consistently account for 60–80% of total BSFL production emissions across published studies (Smetana et al., 2019; Halloran et al., 2016), making grid carbon intensity the single most influential production parameter. This finding carries direct relevance for China: with a coal-dominated electricity grid averaging approximately 0.60 kg CO₂-eq per kWh, Chinese BSFL facilities face a structurally higher emissions burden than the European pilot-scale operations that dominate the published LCA literature. Extrapolating European-derived GWP estimates to Chinese industrial contexts without grid-specific adjustment therefore risks systematically underestimating the climate cost of insect-based feed ingredients. Regarding alternative plant proteins, including rapeseed meal, pea protein, and single-cell protein, these were excluded to focus on waste-valorisation pathways with the highest circular economy potential under Chinese conditions and for which comparative LCA evidence is most critically lacking. Under circular economy principles, livestock fed on food system residuals rather than human-edible crops

can deliver protein with substantially reduced land and resource costs, though the climate performance of specific alternative ingredients remains highly context-dependent (Van Zanten et al., 2018). When properly treated to eliminate pathogens, processed food waste can replace conventional feed ingredients, with environmental benefits extending beyond avoided feed crop production to include credits for prevented landfill methane emissions (Zu Ermgassen et al., 2016).

In Asian contexts, food waste feeding has a long operational history. Japan, South Korea, and China's Taiwan have maintained regulated swill-feeding programs for decades. Saleemdeen et al. (2017), drawing on South Korean heat-treated processing as the working model, found that food waste pig feed outperformed anaerobic digestion and composting across 13 of 14 environmental and health impact categories, with environmental benefits driven primarily by substitution of conventional feed production (Zu Ermgassen et al., 2016). China operated comparable systems at substantial scale prior to 2018, when the emergence of African Swine Fever prompted a nationwide prohibition on raw food waste feeding, subsequently replaced by mandatory thermally treated processed food waste protocols under revised national biosecurity regulations. These thermal treatment requirements, stipulating a minimum of 30 min at 70 °C, introduce measurable additional processing energy burdens that pre-2018 Asian studies did not account for. As a result, earlier GWP estimates for food waste feeding scenarios likely understate current operational emissions under Chinese conditions, while simultaneously the landfill methane avoidance credit remains underexplored in the Chinese waste management context, where methane capture rates at municipal solid waste facilities vary considerably by region and facility type. Taken together, these factors mean that the net climate benefit of food waste feeding under current Chinese regulatory and infrastructural conditions remains genuinely unresolved in the published literature. Moreover, European studies report land-use reductions of 21.5% when food waste replaces conventional pig feed (Zu Ermgassen et al., 2016). Accurately quantifying the net climate performance of food waste feeding under current Chinese biosecurity conditions therefore requires a purpose-built LCA rather than extrapolation from pre-2018 Asian or European literature. Methodologically, this study employs attributional modelling for upstream background processes and consequential system expansion for foreground waste diversion, a distinction elaborated in Section 2.

1.1 Research gaps in current literature

Despite growing interest in alternative protein sources for livestock production, critical knowledge gaps constrain evidence-based decision-making for Chinese pig production systems. First, methodological heterogeneity severely limits comparability across LCA studies, only 38% of published pig production LCAs provide complete feed composition descriptions, and merely 9% report environmental impacts disaggregated by individual feed ingredients (Gislason et al., 2023; Andretta et al., 2021). Without standardized inventory reporting, cross-study comparisons remain unreliable and mitigation conclusions difficult to generalize. Second, existing insect LCA studies focus predominantly on European or laboratory-scale production contexts (Smetana et al., 2019; Halloran et al., 2016), with limited representation of industrial Chinese facilities where energy sources, climatic conditions, waste feedstock composition, and production technologies may differ substantially, differences that can substantially shift GWP estimates depending on grid carbon intensity, rearing

substrate, and production scale (Smetana et al., 2019; Halloran et al., 2016). Third, food waste feeding assessments frequently neglect system expansion credits for avoided waste disposal impacts (Gislason et al., 2023; Andretta et al., 2021), particularly prevented landfill methane emissions, and published studies predate China's intensified biosecurity requirements following African Swine Fever outbreaks (Fan et al., 2023), which mandate thermal treatment protocols that materially alter processing energy requirements and operational logistics. ISO 14044 (cl. 4.3.4) establishes system expansion as the preferred approach to allocation in multi-output processes, yet many studies default to cut-off allocation, assigning zero burden to waste inputs and forfeiting the environmental credit for avoided disposal (ISO 14044:2006, 2006), a methodological choice that can substantially underestimate food waste feeding benefits by up to 30% (Saleemdeen et al., 2017). Fourth, China-specific LCA studies remain scarce; recent assessments report GHG intensities of 3.5–4.5 kg CO₂-eq per kg live weight (Yang et al., 2023; Chen et al., 2025), but comprehensive comparisons of alternative feed formulations within Chinese production contexts accounting for China's coal-intensive electricity grid, domestic transport infrastructure, and locally sourced feedstocks are largely absent from the literature. Fifth, comparative assessments typically evaluate ingredients individually rather than within integrated multi-scenario frameworks, limiting their utility for ranking options or identifying trade-offs between environmental objectives. Sixth, uncertainty propagation and sensitivity analysis remain underexplored; most studies report point estimates without confidence intervals or systematic parameter sensitivity evaluation, making it impossible to determine whether differences between scenarios are statistically meaningful or within the noise of input uncertainty.

This study directly addresses each of these gaps. It adopts a standardized cradle-to-farm-gate LCA framework with full inventory transparency across four feed scenarios, enabling direct comparison. China-specific Ecoinvent datasets and IPCC (2019) Tier 1 manure emission factors replace European proxies wherever data permit. ISO 14044-compliant system expansion is applied to food waste valorization, explicitly quantifying avoided landfill methane credits. All four scenarios are evaluated within a single integrated framework, and Monte Carlo uncertainty propagation across key parameters combined with one-at-a-time sensitivity analysis produces confidence intervals that allow statistically grounded conclusions about scenario rankings.

1.2 Research questions, significance and study overview

The following research questions guided this study:

- 1 RQ1: How do greenhouse gas emissions from Chinese pig production compare across conventional, insect-based, and food waste feeding scenarios when assessed with IPCC (2019) manure calculations and proper food waste system expansion?
- 2 RQ2: What are the relative contributions of feed production, on-farm operations, and manure management to total global warming potential in each scenario?
- 3 RQ3: What do the observed GWP differences between feeding scenarios imply for the feasibility and prioritization of feed-based mitigation strategies in China's pig sector, and how do findings position against published Chinese production benchmarks?

Beyond the research questions above, this study makes three principal contributions to sustainable livestock production knowledge. Methodologically, it demonstrates the quantitative importance of proper LCA accounting through explicit comparison of embedded database values versus region-specific IPCC (2019) calculations for manure emissions, and rigorous application of system expansion for waste valorization credits per ISO 14044 guidelines. Empirically, it provides, to our knowledge, one of the first comprehensive comparisons of insect-based and food waste feeding alternatives within Chinese pig production contexts using corrected methodology and China-specific datasets, addressing a critical geographic gap in existing literature. Practically, results inform evidence-based feed sourcing strategies for China's pig sector, representing 45% of global pork production, with direct relevance to the nation's carbon neutrality commitments by 2060.

The remainder of this paper is organized as follows: Section 2 presents the LCA methodology including goal and scope definition, life cycle inventory development with emphasis on corrected manure emission calculations using IPCC (2019) Guidelines and food waste system expansion implementation, and impact assessment approach. Section 3 reports global warming potential results for all evaluated scenarios with contribution analysis identifying major emission hotspots and benchmarking against published Chinese pig production values. Section 4 discusses findings in relation to existing literature, addresses study limitations, and proposes directions for future research. Section 5 concludes with implications for sustainable feed development and policy recommendations for China's pig sector.

2 Methodology

2.1 Goal and scope definition

The goal of this LCA study is to evaluate and compare the environmental impacts of pig production in China using conventional feed versus alternative feed formulations incorporating black soldier fly larvae (BSFL) meal and processed food waste. Four feeding scenarios were assessed to identify potential environmental benefits and trade-offs associated with replacing conventional protein sources with these alternative ingredients.

The functional unit was defined as 1 kg of pig live weight (LW) at the farm gate. This unit allows direct comparison between scenarios and aligns with standard practice in livestock LCA studies (Yang et al., 2023; Mackenzie et al., 2016). The system boundary followed a cradle-to-farm gate approach, encompassing all upstream processes including feed ingredient production, feed processing and transport, pig housing and management, and manure-related emissions. Post-farm processes such as slaughtering, meat processing, retail, and consumption were excluded from the scope.

2.2 Life cycle inventory development

LCA Software and Database: Life cycle inventory (LCI) modeling was conducted using openLCA 2.5.0 with background data from the Ecoinvent database version 3.10 (Wernet et al., 2016). The cut-off system model was applied, wherein environmental burdens from waste treatment and recycling are assigned to the primary producer of waste materials; consequently, waste inputs entering background

processes carry zero environmental burden. This is complementary to, and consistent with, the system expansion applied in the foreground model: where food waste displaces landfill disposal, avoided landfill methane emissions are credited as a negative emission to air. To avoid burden-shifting while maintaining consistency with international comparative practice, a hybrid attributional-consequential approach was utilised. The cutoff model and system expansion address different life cycle stages and therefore do not overlap: cutoff eliminates the upstream input burden of food waste, assigning the GHG cost of food production to the waste generator rather than the feed user; system expansion credits avoided downstream disposal emissions, quantifying the landfill methane that does not occur because the waste is diverted from landfill. These are distinct environmental flows at distinct system boundaries. In the interest of full transparency, under a pure attributional cutoff approach without the landfill credit, the food waste scenario GWP would be approximately 4.06 kg CO₂-eq/kg LW, marginally above the conventional baseline of 3.9 kg CO₂-eq/kg LW. This confirms that the environmental advantage of food waste feeding is contingent on the avoided landfill emissions credit. The hybrid approach is retained in the primary analysis because ISO 14044 (cl. 4.3.4) identifies system expansion as the preferred method for quantifying avoided waste disposal impacts, and because the landfill methane credit represents a real, measurable physical flow under Chinese municipal waste conditions. China-specific (CN) datasets were used where available in Ecoinvent 3.10 to ensure geographic representativeness, including electricity generation and industrial heat production. Where China-specific datasets were unavailable, Rest of World (RoW) datasets were employed as proxies for crop production (maize grain, soybean meal) and freight transport, representing global average production systems.

- 1 *Feed scenario design*: Four feed scenarios were developed representing different protein sourcing strategies (Supplementary Table S1), formulated to meet nutrient requirements for growing-finishing pigs (National Research Council, 2012). This baseline scenario consisted of a conventional corn-soybean formulation representing standard commercial practice in China. The insect-based scenarios replaced portions of soybean meal with BSFL meal on a protein-equivalent basis at 10 and 20% dietary inclusion levels, yielding a substitution ratio of 0.8 kg soybean meal displaced per 1.0 kg BSFL meal added at 20% inclusion; amino acid and energy requirements per National Research Council (2012) were maintained by adjusting inclusion proportions accordingly. The food waste scenario incorporated 30% processed food waste replacing maize grain (reduced from 68 to 45% of diet DM) and eliminating the mineral fraction (7%), with soybean meal maintained at 25% to preserve dietary protein and amino acid targets. The energy contribution of the reduced maize inclusion was compensated by the carbohydrate and fat fraction of the processed food waste, consistent with its reported proximate composition in the literature. All scenarios were formulated to meet the nutritional requirements of growing-finishing pigs (110 kg final weight) according to National Research Council (2012) guidelines. A constant feed conversion ratio (FCR) of 2.5 kg feed per kg live weight gain was applied across all scenarios to isolate the effects of feed composition from potential differences in animal performance. This conservative assumption may underestimate the benefits

of alternative feeds if they improve feed efficiency but ensures that observed differences in environmental impacts are attributable solely to feed ingredient substitution. It is important to note that while some studies report FCR variation with insect inclusion (Hong and Kim, 2022) a constant FCR was applied to isolate the effects of feed composition on life-cycle impacts. Sensitivity analysis (Section 3.2) examines FCR variation of $\pm 30\%$ to test this assumption's influence on results. The $\pm 30\%$ range was selected to bracket plausible diet-related uncertainty, including potential reductions in digestibility or palatability associated with chitin in BSFL meal at higher inclusion levels (Hong and Kim, 2022) and variation in the energy density and nutrient consistency of processed food waste. The use of a constant FCR is therefore a scenario-control assumption that improves comparability across diets, rather than an empirical claim that animal performance would remain identical under all feeding scenarios. If BSFL inclusion worsens digestibility or intake, the GWP of the BSFL scenarios would likely increase because more feed would be required per kg live weight. For the food waste scenario, reduced nutrient consistency or mineral imbalance could reduce growth efficiency and narrow the estimated climate advantage. The $\pm 30\%$ FCR sensitivity analysis should therefore be interpreted as a bounding test of the model's sensitivity to feed-use efficiency, not as proof of unchanged animal performance across diets. Within the tested range, the scenario ranking remained stable; however, diet-specific feeding trials under Chinese production conditions are needed to validate FCR values for BSFL meal and processed food waste. It should also be noted that the food waste diet omits external mineral supplementation (Supplementary Table S1), with calcium and phosphorus requirements assumed to be partly or fully met by the inherent mineral content of processed food waste. Processed food waste from urban food-service streams may contain calcium and phosphorus from dairy residues, eggshell fragments, fish and meat bones, and cereal-based materials, which have been discussed as nutritionally relevant components in swill-feeding literature (Zu Ermgassen et al., 2016; Saleem et al., 2017). However, independent mineral profiling of the specific waste stream modelled here was not conducted. This assumption should therefore be interpreted as plausible but unverified, and its implications for nutritional equivalence and FCR validity are acknowledged as a study limitation in Section 4.3.

- 2 *Black soldier fly larvae (BSFL) meal production*: BSFL meal production was modeled as a multi-output process based on pilot-scale data from Smetana et al. (2019) and Halloran et al. (2016). Larvae were reared on biowaste substrates (mixed kitchen and garden waste) under controlled conditions. The production system included rearing, handling, drying (to reduce moisture content), and defatting operations to produce a protein concentrate suitable for animal feed. The LCI for BSFL meal production per 1 kg of defatted meal is summarized in Supplementary Table S2. The process generates three co-products: defatted meal (40–60% protein), insect oil (high in medium-chain fatty acids), and frass (nutrient-rich residue suitable as organic fertilizer). Environmental burdens were allocated among co-products using mass allocation as the primary approach (23% to BSFL meal, 7% to insect oil, and 70% to frass by mass yield). Biowaste substrate was treated as a

residual material with zero upstream production burden under the cut-off approach, meaning that emissions from producing the original food or biomass were not assigned to BSFL production. However, foreground operations required to make the substrate usable for larvae, including collection, transport to the rearing facility, sorting, size reduction, and mechanical or thermal pretreatment, can contribute additional emissions. These flows were not fully represented in the pilot-scale inventory sourced from [Smetana et al. \(2019\)](#) and [Halloran et al. \(2016\)](#), and therefore represent a potential source of underestimation for the BSFL scenarios under Chinese commercial conditions. A sensitivity analysis using economic allocation as an alternative showed that the choice of allocation method changed the GWP for the BSFL 20% scenario by approximately $\pm 12\%$, indicating that the conclusions are robust to this methodological choice.

- 3 **Food waste processing:** Food waste processing was modeled to produce sterilized food waste feed suitable for direct pig consumption, following safety regulations for swill feeding where permitted ([Zu Ermgassen et al., 2016](#)). Raw food waste (70% moisture content) undergoes collection, shredding, thermal treatment (70 °C for 30 min to eliminate pathogens), and moisture reduction to produce feed-grade material. The LCI for food waste processing is presented in [Supplementary Table S3](#). The electricity input of 3.0 MJ/kg dry matter represents mechanical operations during food-waste conversion, including shredding or grinding, pumping, mixing, dewatering, and process control. The heat input of 4.0 MJ/kg dry matter represents thermal sanitisation and moisture reduction required to produce feed-grade processed food waste. These values were adopted from published food-waste-to-feed LCI data and applied as average processing inventory values rather than plant-specific engineering measurements. Processing configuration may materially affect this inventory: continuous systems with heat recovery, steady-state operation, and integrated dewatering would likely reduce heat demand per kg dry matter, whereas batch systems with repeated heating and cooling cycles or limited heat recovery may increase thermal losses. Because Chinese plant-level data were unavailable, process configuration remains a source of uncertainty and should be evaluated in future sensitivity analysis. A critical component of this analysis is the application of system expansion to account for avoided landfill emissions. In the absence of use as animal feed, food waste would be disposed of in landfills where anaerobic decomposition generates methane, a potent greenhouse gas. Following [IPCC \(2019\)](#) methodology for solid waste (Chapter 3), avoided methane emissions were calculated based on:
 - a) Degradable organic carbon (DOC) content: 0.15 kg C/kg dry matter (IPCC Table 3.3)
 - b) Methane correction factor (MCF): 1.0 for anaerobic landfill conditions
 - c) DOC dissimilation fraction (DOCf): 1.0 (upper-bound value for food waste, reflecting near-complete anaerobic biodegradability; the [IPCC \(2019\)](#) default of 0.5 applies to mixed MSW, food waste decomposes almost entirely under anaerobic conditions).
 - d) Methane fraction in landfill gas (F): 0.5 (IPCC default)

- e) Landfill gas capture efficiency: 40% (estimated for urban landfills in China)
- f) The calculation yields 0.06 kg CH₄ avoided per kg dry matter food waste ($0.15 \times 1.0 \times 1.0 \times 0.5 \times 16/12 \times 0.6 = 0.060$ kg CH₄), equivalent to 1.68 kg CO₂-eq (using GWP₁₀₀ of 28 for fossil methane per IPCC AR5, consistent with ReCiPe 2016 characterization factors).

Landfill was selected as the system expansion counterfactual because it represents the predominant disposal route for municipal food waste in China, where organised anaerobic digestion and composting infrastructure for separately collected food waste streams remains limited outside designated pilot cities. [Salemdeeb et al. \(2017\)](#) explicitly compared food waste pig feeding against anaerobic digestion and composting across 14 environmental and health impact categories, finding that feeding outperformed both alternatives in 13 of 14 categories. This provides contextual support for the landfill counterfactual and suggests that selecting AD or composting as the counterfactual would reduce rather than increase the estimated benefit of food waste feeding. The exclusion of AD and composting as alternative counterfactuals is acknowledged as a scope limitation in Section 4.3.

- 4 **Feed mixing and transport:** Feed ingredients were transported to a centralized feed mill where mixing operations occurred. Transport distances were based on typical supply chain configurations in China: 300 km for bulk ingredients (maize, soybean meal) from production regions to feed mills, and 80 km for finished feed delivery to farms. Food waste collection involved 40 km average transport distance from urban collection points to processing facilities. All transport was modeled using Rest of World (RoW) freight lorry datasets (7.5–16 metric ton capacity) from Ecoinvent 3.10 due to unavailability of China-specific transport data. Feed mixing energy was estimated at 0.2 MJ electricity per kg finished feed, representing mechanical mixing, conditioning, and pelletization processes, consistent with processing energy values reported for commercial livestock feed manufacture in comparable LCA studies. This is based on standard commercial pelletization energy requirements (0.15–0.25 MJ/kg feed) reported across published livestock LCA studies ([Smetana et al., 2019](#); [Mackenzie et al., 2016](#)).
- 5 **Pig Production System:** Pig production was modeled for growing-finishing pigs from weaning (~25 kg) to market weight (110 kg) over a 140-day period, representing typical commercial production systems in China ([Yang et al., 2023](#)). Housing conditions assumed intensive indoor facilities with liquid slurry manure management systems, which account for approximately 75% of commercial pig production in China ([Food and Agriculture Organization of the United Nations, 2024](#)). Operational inputs per kg live weight gain included:
 - a) Feed: 2.5 kg (scenario-specific composition)
 - b) Electricity: 0.6 MJ (ventilation, lighting, automated systems)
 - c) Heat (natural gas): 0.3 MJ (climate control during cold periods)
 - d) Water: 5.0 kg (drinking water and facility cleaning)

All energy inputs utilized China-specific Ecoinvent datasets to reflect the carbon intensity of China's electricity grid (approximately 0.60 kg CO₂-eq/kWh as characterized in Ecoinvent 3.10) and natural gas infrastructure.

- 6 Manure management emissions: Methane (CH₄) and nitrous oxide (N₂O) emissions from manure management were calculated manually using IPCC (2019) Tier 1 methodology for East Asia climate conditions and liquid/slurry storage systems, rather than relying on Ecoinvent's embedded pig production datasets. This approach ensured accuracy and transparency for China-specific conditions, as embedded datasets may represent global averages not applicable to Chinese production systems. Methane emissions were calculated from IPCC (2019) Table 10.14 annual emission factors for market swine in East Asia (4.0 kg CH₄/head/year for liquid/slurry systems), as shown in Equation (1) below:

$$CH_4 \left(\frac{kg}{kgLW} \right) = \frac{\text{Annual EF} \times \left(\frac{\text{Days on farm}}{365} \right)}{\text{Final body weight}} = \frac{4.0 \times \left(\frac{140}{365} \right)}{110} = 0.014 kgCH_4 \left(\frac{kg}{kgLW} \right) \quad (1)$$

All annual per-head emission factors were converted to per kg live weight at farmgate using the same basis as Equation (1): annual value $\times (140/365) \div 110$ kg final body weight.

Nitrous oxide emissions were calculated accounting for three pathways (IPCC (2019), Tables 10.19, 10.21, 10.22). Direct N₂O from manure storage:

- i N excretion rate: 11 kg N/head/year (IPCC Table 10.19) = 0.0384 kg N/kg LW for 110 kg pig over 140 days
- ii Emission factor (EF₃): 0.002 kg N₂O-N/kg N for liquid/slurry systems
- iii Direct N₂O = 0.0384 $\times$ 0.002 $\times$ (44/28) = 0.000121 kg N₂O/kg LW
- a) Indirect N₂O from volatilization:
 - i Volatilization fraction: 20% of excreted N
 - ii Emission factor (EF₄): 0.01 kg N₂O-N/kg N volatilized
 - iii Indirect N₂O (vol) = 0.0384 $\times$ 0.20 $\times$ 0.01 $\times$ (44/28) = 0.000121 kg N₂O/kg LW
- b) Indirect N₂O from leaching:
 - i Leaching fraction: 30% of excreted N
 - ii Emission factor (EF₅): 0.0075 kg N₂O-N/kg N leached.
 - iii Indirect N₂O (leach) = 0.0384 $\times$ 0.30 $\times$ 0.0075 $\times$ (44/28) = 0.000136 kg N₂O/kg LW

Total N₂O = 0.000378 $\approx$ 0.00040 kg N₂O/kg LW.

These direct emissions (0.014 kg CH₄ and 0.00040 kg N₂O per kg LW) were applied identically across all four scenarios, as feed

composition was not expected to significantly alter manure emission factors under the study's scope. This assumption is conservative; while dietary crude protein content varies (baseline 16.2%, BSFL 20 15.8%, food waste 14.9%), the differences are within typical commercial ranges and unlikely to alter IPCC (2019) Tier 1 emission factors significantly (Rigolot et al., 2010). Sensitivity analysis in Section 3.2 examines $\pm 30\%$ variation in manure CH₄ to bound this uncertainty.

2.3 Life cycle impact assessment

Environmental impacts were assessed using the ReCiPe 2016 Midpoint (H) method (Huijbregts et al., 2017), which incorporates GWP₁₀₀ characterization factors consistent with IPCC AR5 (CH₄ = 28 kg CO₂-eq, N₂O = 265 kg CO₂-eq). Manure and food waste emission factors were derived from the IPCC (2019) Refinement Guidelines as described in Section 2.2, and converted to CO₂-equivalents using these characterization factors (Huijbregts et al., 2017). Six impact categories relevant to livestock production systems were evaluated:

- 1 Global warming potential (GWP): kg CO₂-equivalent, including CO₂, CH₄ (GWP = 28), and N₂O (GWP = 265) over a 100-year timeframe
- 2 Terrestrial acidification: kg SO₂-equivalent
- 3 Freshwater eutrophication: kg P-equivalent
- 4 Marine eutrophication: kg N-equivalent
- 5 Land use: m²·year crop-equivalent
- 6 Water consumption: m³

Impact characterization was performed within openLCA using built-in characterization factors from the ReCiPe 2016 method package. Results were normalized to the functional unit (1 kg pig live weight) for direct comparison across scenarios.

2.4 Data quality and uncertainty

Data quality varied across system components. Foreground data (feed formulations, pig production parameters) were based on standard Chinese commercial practices documented in peer-reviewed literature (Fan et al., 2023; Smetana et al., 2019). Background data from Ecoinvent 3.10 represent global average or regional (China, Brazil) production systems with documented pedigree matrices indicating good-to-fair quality for major agricultural commodities.

BSFL meal production data were sourced from pilot-scale operations in Europe (Smetana et al., 2019; Halloran et al., 2016); no published LCI data from Chinese commercial-scale BSFL facilities currently exists, and the use of European pilot-scale inventories represents the best available proxy. To bound this uncertainty beyond the $\pm 30\%$ range, a supplementary scenario was examined in which BSFL production energy was reduced by 50%, representing an optimistic commercial-scale efficiency gain. Under this scenario, BSFL 20% GWP decreased to approximately 4.6 kg CO₂-eq/kg LW, remaining 18% above baseline. Climate parity with the conventional baseline would require approximately 55% reduction in production energy, a threshold achievable only through substantial renewable energy integration rather than efficiency gains alone. This confirms that the directional finding, BSFL meal increases GWP under current Chinese grid

conditions, is robust to a wide range of production energy assumptions.

Food waste processing parameters were adapted from UK-based swill feeding trials (Zu Ermgassen et al., 2016) with adjustments for thermal treatment requirements and transport distances applicable to Chinese urban contexts. The assumed landfill gas capture efficiency (40%) for system expansion credits represents a moderate estimate between zero capture (older facilities) and 60% capture (modern engineered landfills), with sensitivity testing across this range.

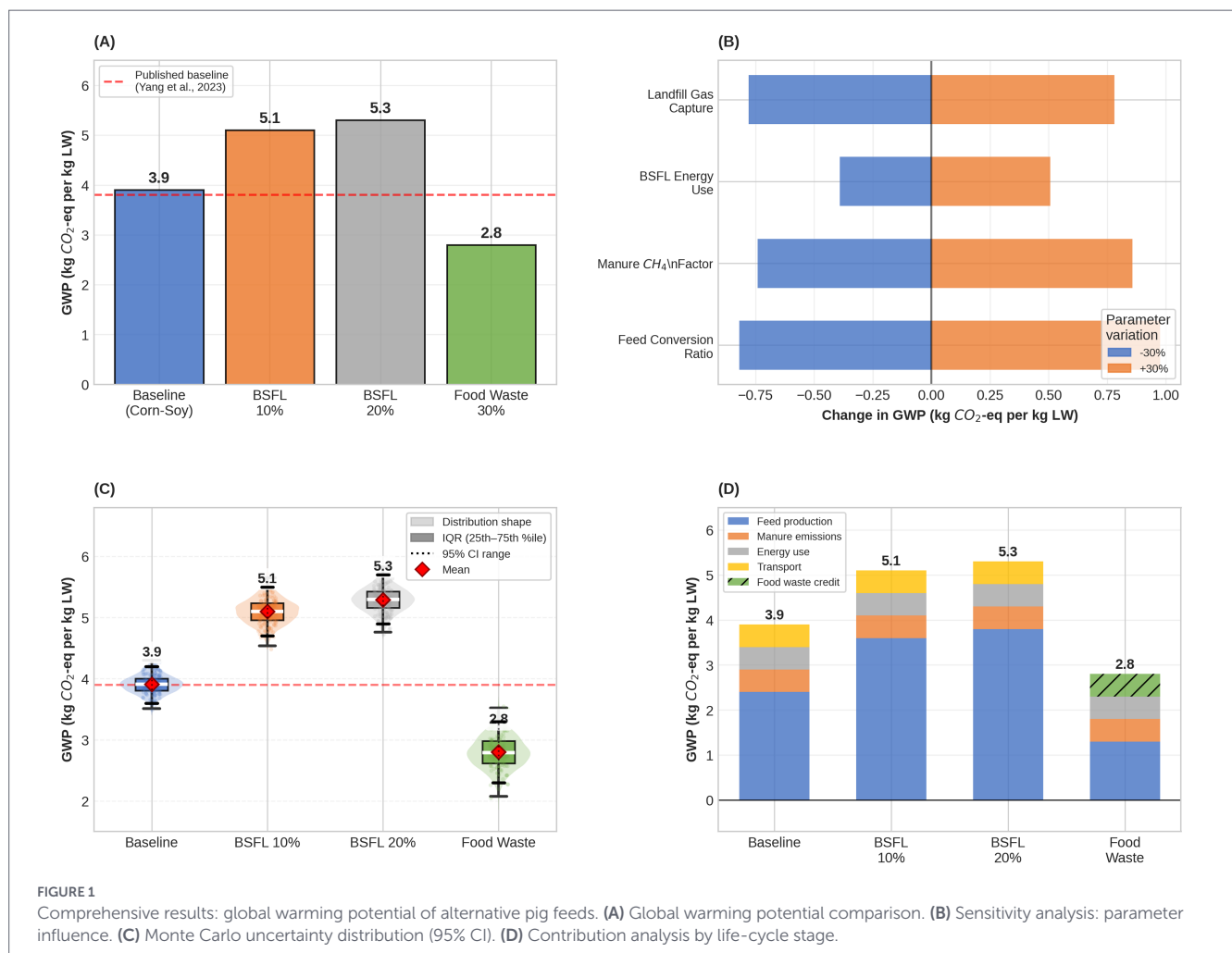
Manure emission factors from IPCC (2019) carry typical uncertainties of $\pm 30\%$ for CH_4 and $\pm 50\%$ for N_2O as noted in the IPCC guidelines. These uncertainties were propagated through Monte Carlo analysis (1,000 iterations) to estimate confidence intervals for final results. Uniform probability distributions were applied to all uncertain parameters, reflecting the absence of empirical distribution data from Chinese pig production systems rather than an assumption of genuine uniformity. While parameters such as CH_4 emission factors and process energy typically follow right-skewed or triangular distributions in practice, uniform distributions produce the widest possible confidence intervals for a given parameter range and therefore represent the most conservative assumption for testing scenario separability. Narrower, more realistic distribution shapes would yield tighter confidence intervals, making the observed non-overlapping 95% CIs across all four scenarios more pronounced rather than less. The choice of uniform distributions

therefore strengthens rather than undermines the robustness of the rank-order conclusions. Future analyses with access to empirical Chinese production data should adopt distribution shapes derived from measured variability.

3 Results

3.1 Life cycle impact assessment results

All emission factors reported below are inventory quantities derived from IPCC (2019) Tier 1 methodology; their conversion to CO_2 -equivalents uses ReCiPe 2016 Midpoint (H) characterisation factors ($\text{CH}_4 = 28$, $\text{N}_2\text{O} = 265$ kg CO_2 -eq) as described in Section 2.3. The global warming potential (GWP) for producing 1 kg pig live weight varied from 2.8 to 5.3 kg CO_2 -eq across the four feed scenarios (Figure 1). The baseline scenario using conventional corn-soybean feed yielded a GWP of 3.9 kg CO_2 -eq/kg LW, closely aligning with recent Chinese pig production LCA studies reporting 3.8 kg CO_2 -eq/kg LW and 3.5–4.5 kg CO_2 -eq/kg LW (Yang et al., 2023; Chen et al., 2025). The insect-based scenarios demonstrated higher GWP compared to baseline, with 5.1 kg CO_2 -eq/kg LW for 10% BSFL inclusion and 5.3 kg CO_2 -eq/kg LW for 20% inclusion. These represent increases of 31 and 36%,



respectively, relative to baseline. The difference between the 10 and 20% BSFL scenarios was not expected to scale linearly with inclusion level because the scenarios represent complete diet reformulations rather than simple additive treatments. Increasing BSFL inclusion also displaced additional soybean meal on a protein-equivalent basis, partly offsetting the added BSFL production burden. In addition, total GWP includes fixed components, particularly manure management and on-farm energy use, which do not change proportionally with BSFL inclusion. Therefore, the 31 and 36% increases should be interpreted as whole-diet scenario outcomes rather than linear dose–response effects of BSFL inclusion. In contrast, the food waste scenario achieved the lowest GWP at 2.8 kg CO₂-eq/kg LW, representing a 28% reduction compared to baseline.

Detailed contribution analysis revealed distinct life-cycle stage dominance patterns across scenarios (Figure 1D); all process-level flows expressed in units consistent with Supplementary Table S3; kg CO₂-eq per kg LW). Feed production contributed 2.4 kg CO₂-eq (62% of total) for the baseline scenario, rising substantially to 3.6 kg CO₂-eq (71%) and 3.8 kg CO₂-eq (72%) for BSFL 10 and 20% scenarios, respectively. This increase reflects the higher energy intensity of BSFL meal production (15 MJ electricity plus 10 MJ heat per kg defatted meal) compared to conventional soybean meal processing (approximately 8 MJ total per kg; Ecoinvent 3.10 database). The BSFL substitution followed nutritional equivalence principles accounting for crude protein content and amino acid digestibility coefficients (van Huis et al., 2013). At 20% BSFL inclusion, the feed formulation displaced 80% of baseline soybean meal on a protein-equivalent basis, resulting in a substitution ratio of 0.8 kg soybean meal replaced per 1.0 kg BSFL meal added. For the food waste scenario, total feed production contributed 3.06 kg CO₂-eq, of which food waste processing (collection, thermal treatment, grinding) accounted for approximately 1.3 kg CO₂-eq, with the remaining 1.76 kg CO₂-eq attributable to the maize and soybean meal still present in the diet (45 and 25% respectively). The avoided emission credit from landfill diversion (−1.26 kg CO₂-eq) assumed baseline landfill methane capture efficiency of 40%, consistent with Chinese municipal solid waste management practices. Sensitivity analysis on this parameter revealed that even under worst-case assumptions (20% capture efficiency), the avoided emission credit remained substantial at −0.50 kg CO₂-eq, maintaining the food waste scenario's environmental advantage over conventional feed albeit with a narrower margin. The food waste environmental advantage is therefore robust in direction across the full sensitivity range tested, but its absolute magnitude is highly dependent on regional landfill gas capture infrastructure, which varies considerably across Chinese municipalities.

Manure management emissions were calculated as 0.50 kg CO₂-eq per kg LW (constant across all scenarios), comprising methane (0.39 kg CO₂-eq from 0.014 kg CH₄) and nitrous oxide (0.11 kg CO₂-eq from 0.00040 kg N₂O). These emissions remained constant across scenarios as feed composition was not expected to substantially alter manure characteristics within the study's scope. The elevated GWP in insect-based scenarios originated primarily from BSFL meal production. At 20% inclusion (0.5 kg BSFL meal consumed per pig at 2.5 FCR), the energy-intensive drying and defatting processes contributed approximately 1.4 kg CO₂-eq, explaining much of the increase over baseline. This burden could be substantially reduced through process optimization, renewable energy integration, or utilization of

waste heat from other industrial processes. The food waste scenario achieved GWP reduction through two mechanisms:

- 1 Partial replacement of energy-intensive maize production with low-burden food waste feedstock.
- 2 Avoided landfill methane emissions credited through system expansion.

The food waste credit of −1.26 kg CO₂-eq derives from 0.75 kg dry matter food waste consumed per kg LW (30% DM inclusion × 2.5 FCR). At 70% moisture content, this corresponds to approximately 2.5 kg wet weight food waste per kg pig live weight.

3.2 Comparison with published benchmarks

The study's baseline GWP of 3.9 kg CO₂-eq/kg LW aligns closely with recent Chinese pig LCA studies: Yang et al. (2023) reviewed Chinese pig production carbon footprints reporting values consistent with our baseline, while Chen et al. (2025) found a range of 3.5–4.5 kg CO₂-eq depending on regional production systems. This correspondence falls within the range reported by recent Chinese pig LCA studies and confirms the representativeness of our baseline scenario for commercial Chinese pork production. The food waste scenario's 28% GWP reduction is consistent with Zu Ermgassen et al. (2016), who reported 21.5% land use reduction and substantial climate benefits from swill feeding in European contexts (Zu Ermgassen et al., 2016). The study's finding of increased GWP for insect-based feeds aligns with recent meta-analyses noting that pilot-scale insect production can be energy-intensive (Smetana et al., 2019; Halloran et al., 2016), though commercial-scale operations with renewable energy may approach or exceed the environmental performance of conventional feeds.

3.3 Sensitivity and uncertainty analysis

Sensitivity analysis examined the influence of key parameters on GWP results:

- 1 Feed conversion ratio (FCR): Varying FCR ± 30% (1.75–3.25 kg feed/kg LW) produced asymmetric GWP changes of −21 to +25% relative to the central estimate. This asymmetry arises because manure management emissions (0.50 kg CO₂-eq/kg LW) were held constant across the FCR range, reflecting their dependence on animal physiology rather than feed quantity, while only feed-related emissions scaled proportionally with FCR. The asymmetry therefore confirms that feed production dominates the GWP profile but does not account for it entirely, and that efficiency gains deliver somewhat smaller proportional benefits than efficiency losses impose penalties.
- 2 Manure CH₄ emission factor: Adjusting the IPCC emission factor ±30% (2.8–5.2 kg CH₄/head/year) altered GWP by −19 to +22%, underscoring the importance of manure management strategies and accurate emission monitoring.
- 3 BSFL production energy: Reducing electricity and heat inputs by 30% (representing potential commercial-scale efficiency gains or renewable energy integration) decreased insect scenario GWP by 10–11%, while a 30% increase raised GWP by 11–13%.

- 4 Landfill gas capture efficiency: Testing 0–60% capture rates for the food waste credit yielded a $\pm 40\%$ range in the food waste scenario GWP (2.3–3.3 kg CO₂-eq), highlighting this assumption's critical influence on waste-based feed conclusions.

Monte Carlo simulation (1,000 iterations) propagating uncertainty in manure emissions ($\pm 30\%$ for CH₄, $\pm 50\%$ for N₂O), BSFL energy use ($\pm 30\%$), and food waste credit (landfill gas capture 20–60%) yielded 95% confidence intervals of:

- 1 Baseline: 3.6–4.2 kg CO₂-eq/kg LW
- 2 Insect 10%: 4.7–5.5 kg CO₂-eq/kg LW
- 3 Insect 20%: 4.9–5.7 kg CO₂-eq/kg LW
- 4 Food waste: 2.3–3.3 kg CO₂-eq/kg LW

Despite uncertainty ranges, all conclusions remained robust: insect-based feeds consistently increased GWP, while food waste feeding consistently reduced GWP relative to baseline across the full uncertainty space.

4 Discussion

The following discussion interprets these findings within the constraints of current Chinese regulatory frameworks and economic feasibility, recognising that environmental performance alone does not determine adoption trajectories.

4.1 Regional context and comparability

The study's baseline GWP of 3.9 kg CO₂-eq/kg LW aligns with published Chinese estimates (Yang et al., 2023) but remains higher than reported European averages (Nguyen et al., 2012; Mackenzie et al., 2016). This difference primarily reflects China's coal-dominated electricity grid composition, longer average feed transport distances, and production system characteristics. The BSFL production energy requirements assumed in this study derive from available literature (Smetana et al., 2019; Halloran et al., 2016); actual Chinese industrial facilities may demonstrate different energy intensities depending on technology adoption, scale, and regional energy infrastructure. Similarly, food waste collection efficiency and processing infrastructure maturity varies substantially across Chinese provinces, affecting the practical applicability and environmental performance of food waste feeding scenarios.

The central finding of this study is that feed ingredient choice drives sharply divergent climate outcomes in Chinese pig production, and that the direction of that divergence is not predictable from ingredient properties alone. Food waste feeding reduced GWP by 28% below baseline through two compounding mechanisms avoided upstream maize production and a landfill methane avoidance credit of -1.26 kg CO₂-eq per functional unit while BSFL meal increased GWP by 31–36% above baseline due to the energy intensity of pilot-scale drying and defatting, a penalty that persisted even after economic co-product allocation following ISO 14044 guidelines (ISO 14044:2006, 2006). The fact that manure emissions were constant across all scenarios at 0.50 kg CO₂-eq/kg LW is consistent with these divergent outcomes being attributable primarily to upstream feed production, a finding aligned with the broader livestock LCA literature

identifying feed as the dominant emission hotspot (Poore and Nemecek, 2018; Yang et al., 2023). This has a direct methodological implication: studies that aggregate feed and on-farm contributions without ingredient-level disaggregation will systematically obscure the leverage points available to producers (Andretta et al., 2021).

These results are broadly consistent with existing literature while resolving ambiguities it leaves open. The baseline GWP of 3.9 kg CO₂-eq/kg LW falls within the range of 3.5–4.5 kg CO₂-eq/kg LW reported by Yang et al. (2023) and Chen et al. (2025) for commercial Chinese production systems, providing confidence in inventory representativeness. The food waste benefit aligns directionally with Zu Ermgassen et al. (2016), who documented substantial climate and land-use gains from swill feeding in European contexts, and extends that finding by quantifying the landfill methane credit explicitly under IPCC (2019) methodology (IPCC, 2019), a component frequently omitted in prior assessments. The elevated GWP for BSFL meal is consistent with Halloran et al. (2016) and Smetana et al. (2019), whose pilot-scale inventories underpin the present analysis, and corroborates the broader finding that insect production energy intensity is the critical variable.

However, this study should not be read as a general verdict against insect-based feeds. Van Zanten et al. (2018) demonstrate that under circular economy conditions, genuine food waste substrates and renewable electricity; BSFL systems can outperform conventional soy on climate metrics. Our results reflect pilot-scale production under China's current grid intensity of approximately 0.58 kg CO₂-eq/kWh as characterised in Ecoinvent 3.10 (Wernet et al., 2016); a 30% reduction in BSFL production energy narrows the gap with baseline by only 10–11%, indicating that grid decarbonisation matters more than production efficiency gains alone. The conditions under which BSFL becomes climate-competitive are therefore specific and achievable, but not yet present at scale in China.

Two limitations warrant quantification rather than acknowledgement alone. First, the BSFL inventory was derived from European pilot-scale operations (Smetana et al., 2019; Halloran et al., 2016) rather than Chinese commercial facilities; no such dataset currently exists. Sensitivity analysis shows that even a 30% reduction in production energy leaves the BSFL 20% scenario 21% above baseline under current grid conditions, suggesting that full climate competitiveness requires either substantial grid decarbonisation or simultaneous gains in both energy efficiency and renewable sourcing. Second, the food waste system expansion credit is acutely sensitive to landfill gas capture efficiency, calculated here following IPCC (2019) solid waste methodology. Testing 0–60% capture rates produced a GWP range of 2.3–3.3 kg CO₂-eq/kg LW for the food waste scenario, a spread larger than the gap between the baseline and food waste central estimates, reflecting genuine variability in Chinese municipal waste infrastructure rather than modelling imprecision. Despite both uncertainties, Monte Carlo analysis yielded non-overlapping 95% confidence intervals across all four scenarios, confirming that the rank order of results is robust to the full parameter space tested.

The practical implications differ substantially between ingredient types. Food waste feeding represents a potentially near-term deployable mitigation strategy requiring no new technology and delivering reductions robust across the full uncertainty range tested, with direct relevance to China's carbon neutrality commitments by 2060, provided that harmonized biosecurity regulation is in place (Fan et al., 2023; Chen et al., 2021). Its scaling, however, is contingent on regulatory conditions: following African Swine Fever outbreaks from 2018

onwards, China mandated thermal treatment above 70 °C for 30 min and provincial licensing for food waste feeding (Fan et al., 2023), meaning widespread adoption requires harmonized national standards rather than farm-level initiative alone. BSFL meal presents a different trajectory, the GWP penalty observed here is not inherent to the ingredient but specific to pilot-scale processing under grid electricity (Smetana et al., 2019; Halloran et al., 2016). Policymakers seeking to accelerate the transition should prioritize renewable energy incentives and waste-heat co-location mandates for insect feed facilities, which would target the drying and defatting operations identified here as the dominant emission source directly.

Beyond the substantive findings, this study demonstrates that methodological choices in livestock LCA are not neutral. Applying region-specific (IPCC, 2019) manure emission factors rather than Ecoinvent embedded values, and implementing ISO 14044-compliant (ISO 14044:2006, 2006) system expansion rather than cut-off allocation for food waste, materially altered GWP outcomes in ways that would have changed the ranking of scenarios under less rigorous accounting, consistent with concerns raised by Andretta et al. (2021) regarding methodological heterogeneity in pig production LCA. These choices should be adopted as minimum reporting standards in livestock feed substitution studies. Three research priorities follow directly from the limitations identified above: development of China-specific BSFL production inventories to replace the European pilot-scale proxy data used here; empirical measurement of landfill gas capture efficiency at Chinese urban waste facilities to resolve the dominant uncertainty in food waste scenario outcomes; and controlled feeding trials reporting scenario-specific feed conversion ratios for BSFL and food waste diets, to capture the asymmetric risk that FCR deterioration at high insect inclusion imposes a larger GWP penalty than an equivalent efficiency improvement delivers as a benefit.

4.2 Policy implications and implementation considerations

The environmental performance differences identified here must be evaluated alongside regulatory frameworks and economic feasibility when considering practical implementation. Current Chinese regulations permit insect protein use in aquaculture feeds but maintain restrictions on pig feed applications pending additional safety evaluations and production standardization. This regulatory barrier persists despite BSFLs demonstrated nutritional adequacy and established use in other jurisdictions, suggesting that policy advocacy and stakeholder engagement remain necessary prerequisites for commercial adoption regardless of environmental performance. For food waste feeding, China's post-African Swine Fever biosecurity requirements mandate thermal treatment (≥ 70 °C for ≥ 30 min) and regular pathogen monitoring, adding operational costs estimated at ¥150–200 per tonne processed feed. These compliance costs may offset economic advantages relative to conventional ingredients, potentially limiting adoption to larger-scale operations with sufficient capital for processing infrastructure investment. However, policy instruments could improve viability: waste processing subsidies under circular economy initiatives, feed ingredient diversification support programs, or integration with existing municipal organic waste management systems could reduce costs while advancing multiple policy objectives simultaneously (Jurgilevich et al., 2016). The constant manure emission contribution across all scenarios (0.50 kg CO₂-eq/kg LW, representing

13% of total GWP) underscores that dietary interventions alone cannot fully address livestock greenhouse gas emissions. Complementary manure management strategies warrant investigation, including anaerobic digestion for biogas recovery, aerobic composting systems, or integration with crop production systems for nutrient recycling. Such integrated approaches could address methane and nitrous oxide emissions while generating renewable energy co-products or displacing synthetic fertilizer production, potentially achieving additional environmental benefits beyond feed ingredient substitution alone.

4.3 Study limitations

This study has several limitations that warrant acknowledgment. First, BSFL production inventory data derive exclusively from European pilot-scale studies; no Chinese commercial-scale LCI dataset currently exists. Substrate composition, climatic control requirements, and operational scale in Chinese facilities may differ from the European baseline, and the magnitude of the GWP increase for BSFL scenarios should be interpreted with this provenance uncertainty in mind, even though the directional finding is robust across the full sensitivity range tested. Additionally, the BSFL production inventory does not fully represent China-specific substrate collection, sorting, transport, and pretreatment. Including these additional foreground burdens would likely increase the GWP of BSFL meal unless offset by avoided waste-management credits or low-carbon processing energy; therefore, the BSFL results reported here may be conservative underestimates. Second, we applied uniform probability distributions for uncertainty analysis; empirical distributions derived from Chinese production data would improve precision and representativeness. Third, our system boundary excluded capital equipment and infrastructure; while typically contributing less than 5% of total environmental impacts in agricultural LCA studies, their inclusion would provide completeness. Fourth, we focused exclusively on global warming potential; comprehensive sustainability assessment requires evaluation of trade-offs across multiple impact categories including eutrophication potential, land use, water consumption, and biodiversity impacts (Poore and Nemecek, 2018). Fifth, the assumed mineral adequacy of the processed food waste in the absence of external supplementation remains an assumption requiring empirical validation, and any associated FCR fluctuations were outside the scope of this study. The constant FCR assumption should be interpreted as a scenario-control device rather than an empirical claim of identical animal performance across diets; diet-specific feeding trials are needed to confirm whether BSFL meal and processed food waste maintain comparable intake, digestibility, and growth efficiency under Chinese production conditions. Sixth, the system expansion credited only avoided landfill disposal as the counterfactual for food waste diversion; anaerobic digestion with biogas recovery and composting were outside the study scope. In contexts where AD infrastructure is more developed, the marginal counterfactual may differ from landfill, which could reduce the magnitude of the food waste feeding credit relative to the values reported here.

4.4 Future research priorities

Several research priorities emerge from this analysis that warrant future investigation. First, primary data collection from operational

Chinese BSFL production facilities would reduce reliance on European-derived parameters and improve representativeness for Chinese conditions. Specific data requirements include energy consumption patterns under different climate zones, waste substrate conversion efficiencies for locally available feedstocks, and co-product yield distributions from industrial-scale operations. Second, economic life cycle assessment or life cycle costing (LCC) integration would provide multi-dimensional sustainability evaluation, informing economically viable mitigation pathways (Swarr et al., 2011). Price volatility for conventional feed ingredients, BSFL production economies of scale, and waste processing infrastructure investment requirements all influence practical adoption potential beyond environmental performance alone. Third, social life cycle assessment dimensions merit exploration within China's socioeconomic context, including implications for smallholder farmer livelihoods, rural employment generation from waste processing facilities, and consumer acceptance of animal products from alternative-fed livestock. Fourth, whole-farm system optimization through integrated manure management LCA would address the 13% GWP contribution from manure emissions identified across all scenarios. Anaerobic digestion modeling, composting systems assessment, and nutrient recycling integration with crop production warrant investigation as complementary strategies (Zhang et al., 2013). Finally, scaling analyses examining provincial-level or national-level feed substitution scenarios could inform policy planning, quantifying aggregate environmental implications and resource requirements for large-scale alternative ingredient adoption aligned with China's 14th Five-Year Plan objectives for agricultural sustainability and food security.

5 Conclusion

Feed production dominates the greenhouse gas footprint of pig production, yet rigorous China-specific evidence comparing alternative ingredients has remained scarce. This study addressed that gap through a comparative LCA of four feeding scenarios, applying IPCC (2019) manure emission factors and ISO 14044-compliant system expansion to ensure methodological choices did not obscure differences between ingredient types. The results reveal a consequential divergence. Food waste feeding reduced GWP by 28% to 2.8 kg CO₂-eq/kg LW, driven by avoided maize production and a landfill methane avoidance credit of −1.26 kg CO₂-eq per functional unit. BSFL meal increased GWP by 31–36% above the 3.9 kg CO₂-eq/kg LW baseline due to the energy intensity of pilot-scale processing, a penalty robust across Monte Carlo uncertainty analysis. Constant manure emissions across all scenarios are consistent with the divergence being driven by upstream feed production, suggesting that ingredient substitution is a meaningful lever for mitigation. For China's path to carbon neutrality by 2060, implications differ by ingredient. Food waste feeding is deployable within existing infrastructure but depends on harmonized biosecurity regulation following African Swine Fever restrictions.

The BSFL penalty is not inherent to the ingredient but specific to pilot-scale processing under grid electricity; renewable energy incentives and waste-heat co-location mandates would target its dominant source directly. Three limitations define the research agenda: reliance on European BSFL inventories requires replacement with China-specific commercial-scale data; food waste credit

uncertainty requires empirical measurement of landfill gas capture efficiency at Chinese facilities; and the constant FCR assumption requires controlled feeding trials to capture the full performance consequences of ingredient substitution. Circular economy feed approaches can deliver measurable GWP reductions under the conditions modelled here. Whether they scale depends on decarbonizing insect processing, harmonizing food waste regulation, and developing China-specific production inventories. Each is tractable.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary material](#), further inquiries can be directed to the corresponding author.

Author contributions

BH: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. ZH: Funding acquisition, Supervision, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fsufs.2026.1841174/full#supplementary-material>

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